

Exercise sheet 05 - Computational Statistical Physics

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Task 1. Implementation of the molecular dynamics simulation.

We consider a system of $N = 40$ identical classical particles in a three-dimensional cubic box of side length $L = 10$, subject to periodic boundary conditions and interacting through the Lennard–Jones potential. The dynamics are integrated with the Leapfrog scheme.

(a) Relative position in periodic boundary conditions.

The relative displacement between two particles i and j is defined as

$$\mathbf{r}_{ij} = \mathbf{r}_i - \mathbf{r}_j.$$

Since the system is simulated in a periodic box, this displacement must be computed using the minimum-image convention. Starting from the naive displacement vector, each Cartesian component is shifted by $\pm L$ whenever necessary, so that the final component lies in the interval $[-L/2, L/2]$. In this way, the interaction between two particles is always evaluated using the shortest periodic image distance.

(b) Leapfrog integration scheme with periodic boundary conditions.

The time evolution of the system is performed with the Leapfrog method. In this scheme, the particle positions are defined at integer time steps, while the velocities are defined at half-integer time steps. At each iteration, the current positions are first wrapped back into the simulation box to enforce periodic boundary conditions. Then, the forces are computed from the particle positions at time t , and the half-step velocities are updated according to

$$\mathbf{v}\left(t + \frac{\Delta t}{2}\right) = \mathbf{v}\left(t - \frac{\Delta t}{2}\right) + \mathbf{a}(t)\Delta t.$$

Since the particle mass is taken to be $m = 1$, the acceleration is equal to the total force. The new particle positions are then obtained from

$$\mathbf{r}(t + \Delta t) = \mathbf{r}(t) + \mathbf{v}\left(t + \frac{\Delta t}{2}\right)\Delta t.$$

After the position update, periodic boundary conditions are applied again so that all particles remain inside the box.

(c) Velocity at integer times.

In the Leapfrog scheme, the velocities are naturally defined at half time steps. To estimate the velocity at integer times, we use the average of the neighbouring half-step velocities,

$$\mathbf{v}(t) = \frac{\mathbf{v}\left(t + \frac{\Delta t}{2}\right) + \mathbf{v}\left(t - \frac{\Delta t}{2}\right)}{2}.$$

This estimate is used, for example, in the computation of the kinetic energy. This averaging is particularly important in a periodic system, since naive finite differences of positions can produce artificial spikes in the

kinetic energy whenever particles cross the boundaries of the box.

(d) Cut-off radius for the potential and the force.

To reduce the computational cost, a distance cut-off $r_c = 2.5$ is introduced in the evaluation of both the Lennard–Jones potential and the corresponding force. Thus, for any particle pair with separation $r \geq r_c$, the interaction is neglected and both the pair potential and the pair force are set to zero. This is justified because the Lennard–Jones interaction decays rapidly with distance, so contributions from sufficiently distant particles are negligible while still requiring computational effort.

(e) Estimate of the maximum number of particles.

Following the approximation suggested in the assignment, we interpret $r_c/2$ as an effective particle radius. Therefore, each particle has an effective diameter r_c . Under a simple-cubic packing assumption, the number of particles that can fit along one side of the box is approximately

$$\left\lfloor \frac{L}{r_c} \right\rfloor.$$

Hence, the maximum number of particles that fit in the full three-dimensional box is

$$N_{\max} = \left\lfloor \frac{L}{r_c} \right\rfloor^3.$$

For the present choice $L = 10$ and $r_c = 2.5$, this gives

$$N_{\max} = \left\lfloor \frac{10}{2.5} \right\rfloor^3 = 4^3 = 64.$$

(f) Initial conditions.

The initial conditions must be chosen carefully in order to avoid unphysical overlaps between particles, which would lead to extremely large repulsive forces and numerical instabilities at the beginning of the simulation. For this reason, the initial particle positions are placed on a regular cubic lattice inside the simulation box. This ensures that the particles are initially well separated. The initial velocities are sampled from a three-dimensional Gaussian distribution. In addition, the mean velocity is subtracted from all particles so that the total initial momentum of the system vanishes and the center of mass does not drift. Finally, the initial half-step velocities required by the Leapfrog scheme are constructed from the initial velocities and the initial forces.

Task 2. Time evolution of the total energy and its fluctuations.

(a) Total energy as a function of time.

To assess the stability of the molecular dynamics simulation, we computed the total energy

$$E(t) = E_{\text{kin}}(t) + E_{\text{pot}}(t)$$

at every time step and plotted it as a function of time. In the chosen units, the particle mass was taken to be $m = 1$, so the kinetic energy was evaluated as

$$E_{\text{kin}}(t) = \frac{1}{2} \sum_{i=1}^N |\mathbf{v}_i(t)|^2,$$

while the potential energy was obtained by summing the Lennard–Jones interaction over all distinct particle pairs.

The resulting energy plot shows that the total energy remains close to a nearly constant value throughout the simulation, apart from a noticeable initial transient. More specifically, the initial energy was

$$E(0) = 50.7398,$$

while the mean energy over the full trajectory was

$$\langle E \rangle = 49.9150.$$

The standard deviation of the energy was found to be

$$\text{Std}(E) = 0.0653,$$

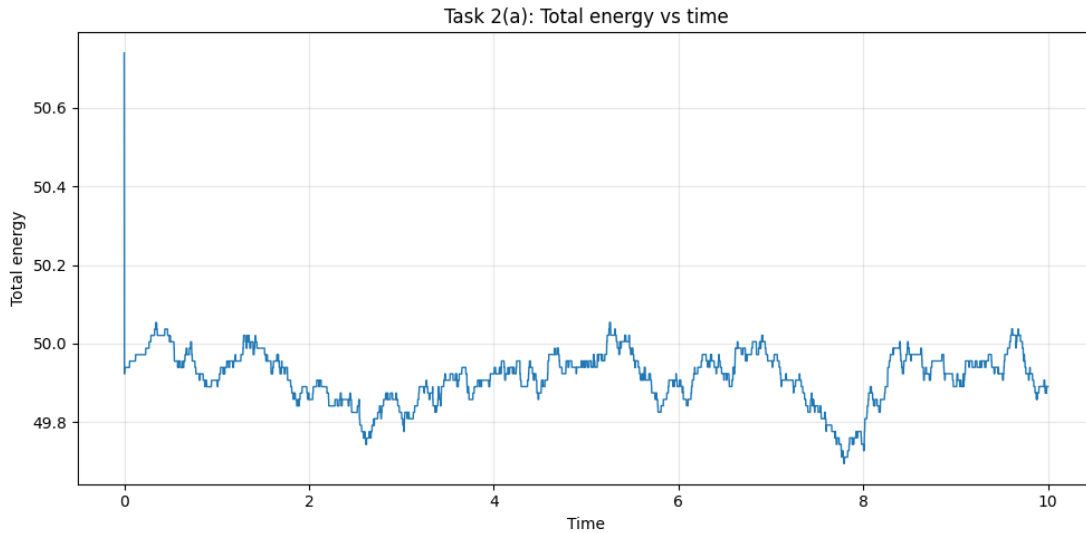
and the maximum absolute deviation from the initial value was

$$\max |E(t) - E(0)| = 1.0451,$$

corresponding to a relative maximum drift of approximately

$$2.06\%.$$

From the plot, one observes that the largest deviation occurs at the beginning of the simulation, whereas afterwards the total energy oscillates in a relatively narrow band around its average value, roughly between 49.8 and 50.0. Therefore, the simulation does not exhibit a systematic long-time increase or decrease of the energy. This indicates that the Leapfrog integration scheme is stable for the chosen time step and that the total energy is approximately conserved, with the main discrepancy concentrated in the initial relaxation stage.



Σχήμα 1: Task 2(a): total energy as a function of time. A pronounced initial transient is visible, after which the energy remains bounded and oscillates around an approximately constant value.

(b) Energy fluctuations.

To analyse the deviations from the average energy more clearly, we defined the energy fluctuation as

$$\Delta E(t) = E(t) - \langle E \rangle.$$

By construction, its time average is zero, which is confirmed numerically by

$$\langle \Delta E \rangle \approx -2.36 \times 10^{-15},$$

i.e. essentially zero up to machine precision.

The fluctuation plot shows that, after the initial transient, the energy oscillates around zero without any visible secular trend. The standard deviation of the fluctuations is

$$\text{Std}(\Delta E) = 0.0653,$$

while the maximum absolute fluctuation is

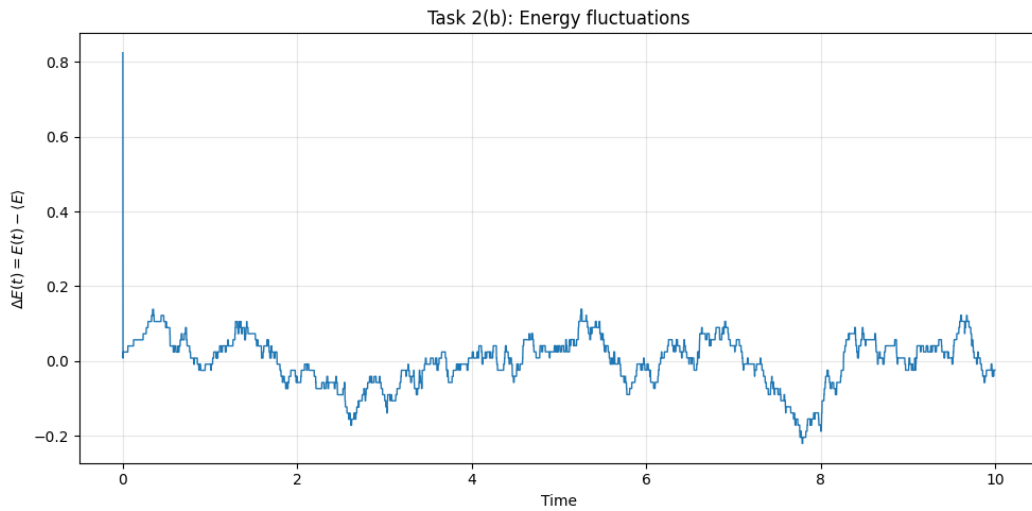
$$\max |\Delta E(t)| = 0.8248.$$

The corresponding relative fluctuation plot,

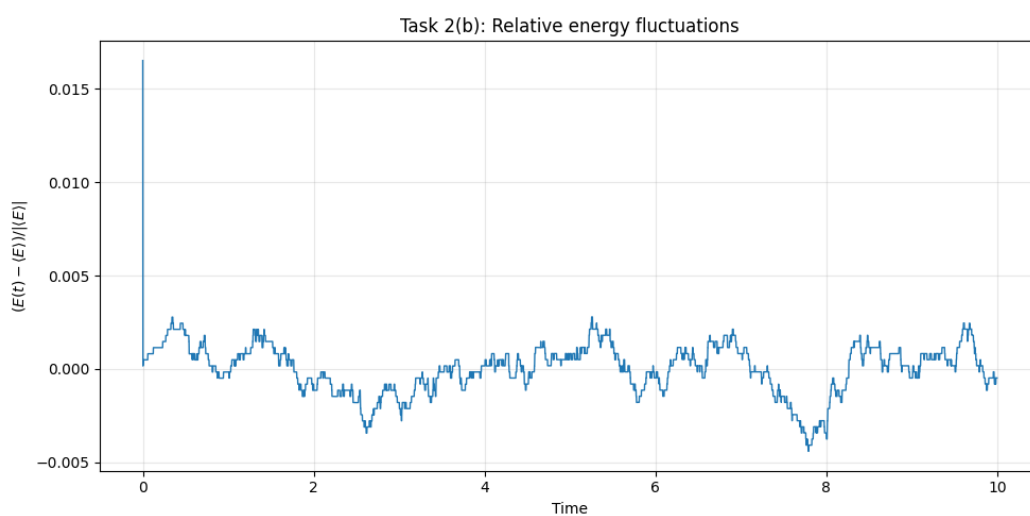
$$\frac{E(t) - \langle E \rangle}{|\langle E \rangle|},$$

shows that the typical relative deviations remain small, of the order of 10^{-3} , apart from the initial peak.

Overall, the results of Task 2 show that the implemented molecular dynamics simulation is numerically stable. The total energy remains bounded and approximately conserved over time, while the energy fluctuations stay centered around zero and do not grow systematically. The dominant discrepancy is the initial peak, which can be attributed to the adjustment of the system from the chosen initial conditions to its subsequent dynamical evolution.



Σχήμα 2: Task 2(b): absolute energy fluctuations, $\Delta E(t) = E(t) - \langle E \rangle$. The fluctuations remain bounded and oscillate around zero after the initial transient.



Σχήμα 3: Task 2(b): relative energy fluctuations, $(E(t) - \langle E \rangle) / |\langle E \rangle|$. The relative deviations remain small, typically of order 10^{-3} , except for the initial peak.

Task 3. Influence of the time discretisation size.

In order to study the influence of the time step Δt on the numerical behaviour of the simulation, we repeated the molecular dynamics run for several different values of Δt , namely

$$\Delta t = 5 \times 10^{-4}, \quad 10^{-3}, \quad 1.5 \times 10^{-3}, \quad 2 \times 10^{-3},$$

while keeping all other physical parameters fixed. For each case, we computed the total energy and then evaluated the energy fluctuations

$$\Delta E(t) = E(t) - \langle E \rangle,$$

as well as the corresponding relative fluctuations

$$\frac{E(t) - \langle E \rangle}{|\langle E \rangle|}.$$

The resulting plots show that the overall behaviour is qualitatively similar for all tested values of the time step. In all cases, the fluctuations remain bounded and oscillate around zero, without any clear secular drift. This indicates that the Leapfrog integrator remains stable in the tested range of Δt . A pronounced initial peak is present in all curves, which is associated with the initial relaxation of the system from the chosen initial conditions.

To quantify the dependence on the time step, we compared the standard deviation and the maximum absolute value of the fluctuations. The corresponding values are

Δt	Std(ΔE)	max $ \Delta E $	Std($\frac{\Delta E}{ E }$)	max $ \frac{\Delta E}{ E } $
0.0005	7.5896×10^{-2}	8.2549×10^{-1}	1.5205×10^{-3}	1.6538×10^{-2}
0.0010	6.5313×10^{-2}	8.2477×10^{-1}	1.3085×10^{-3}	1.6523×10^{-2}
0.0015	6.7946×10^{-2}	8.4016×10^{-1}	1.3617×10^{-3}	1.6837×10^{-2}
0.0020	7.1297×10^{-2}	8.3522×10^{-1}	1.4287×10^{-3}	1.6736×10^{-2}

From these results, we observe that the smallest standard deviation of the fluctuations is obtained for

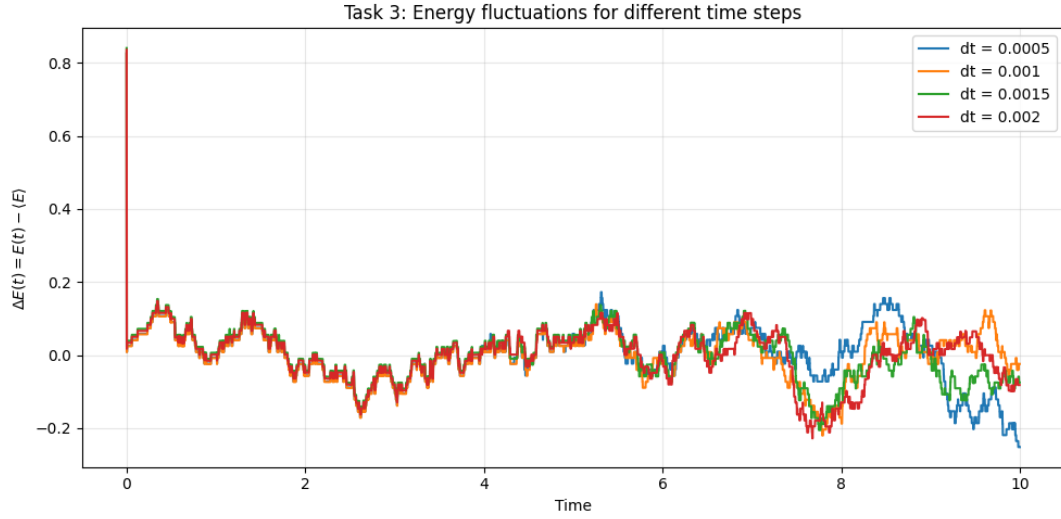
$$\Delta t = 10^{-3},$$

for which both the absolute and relative fluctuation measures are minimal among the tested values. For larger time steps, such as $\Delta t = 1.5 \times 10^{-3}$ and $\Delta t = 2 \times 10^{-3}$, the fluctuations become slightly larger, which is consistent with the expected increase of discretisation error. Interestingly, the smallest tested value, $\Delta t = 5 \times 10^{-4}$, does not produce the smallest fluctuation level in this particular simulation. This suggests that, within the present finite simulation time and for the chosen initial conditions, statistical variability and the initial transient also affect the quantitative comparison.

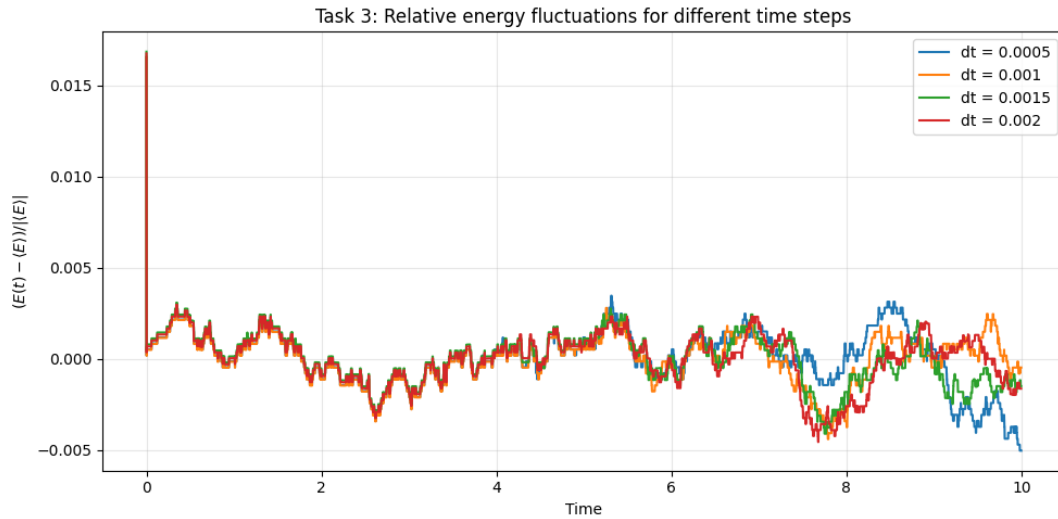
Overall, the comparison confirms that the time step has a visible influence on the accuracy of the simulation, although all tested values remain numerically stable. Among the values considered here, $\Delta t = 10^{-3}$ provides the best balance between numerical accuracy and computational efficiency, since it yields the smallest fluctuation level while avoiding the additional computational cost of a smaller time step.

Δt	$\text{Std}(\Delta E)$	$\max \Delta E $	$\text{Std}(\Delta E/ E)$	$\max \Delta E/ E $
0.0005	7.5896×10^{-2}	8.2549×10^{-1}	1.5205×10^{-3}	1.6538×10^{-2}
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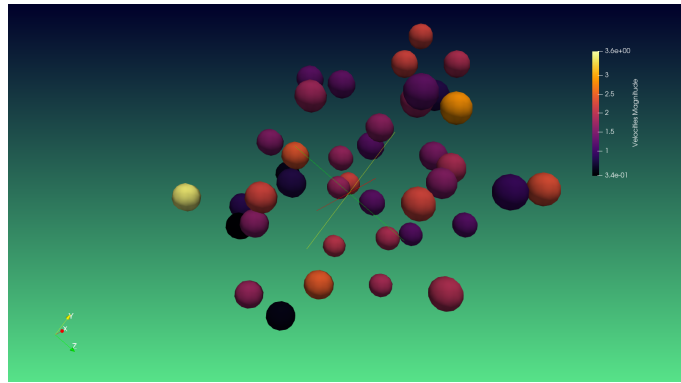
Πίνακας 1: Quantitative comparison of the absolute and relative energy fluctuations for the different time steps.



Σχήμα 4: Task 3: absolute energy fluctuations for different values of Δt . All tested time steps produce bounded fluctuations with qualitatively similar behaviour.



Σχήμα 5: Task 3: relative energy fluctuations for different values of Δt . The case $\Delta t = 10^{-3}$ gives the smallest fluctuation level among the tested values.



Σχήμα 6: Paraview Simulation

Visualisation of the simulation.

In addition to the energy-based analysis, we visualised the particle dynamics with ParaView. For this purpose, the particle coordinates were exported at multiple time steps in VTK unstructured-grid format, together with additional point data containing the particle velocities, forces, radii, and scalar values used for colouring. This is consistent with the visualisation workflow suggested in the exercise sheet, where the simulation output is brought into a ParaView-compatible format using the provided `vtktools.py` script. The exported dataset contains the particle positions in the `Points` array, as well as vector fields associated with the particles, such as velocities and forces.

The generated snapshots were grouped into a `.pvd` collection file, which was then loaded into ParaView. The particle positions were displayed using the Glyph filter with Glyph Type = Sphere, so that each particle appears as a sphere in the three-dimensional simulation box. The particles were coloured according to the magnitude of their velocity, which allows a direct visual identification of faster and slower particles during the time evolution.

To visualise the interaction forces, a second Glyph filter was applied to the same dataset, this time using Glyph Type = Arrow. The arrows represent the instantaneous forces acting on the particles, and their direction and size encode the corresponding force vectors. In this way, the ParaView representation makes it possible to inspect not only the particle motion itself, but also the instantaneous mechanical interactions responsible for that motion.

A box representing the simulation domain was added with dimensions

$$10 \times 10 \times 10$$

and center at

$$(5, 5, 5),$$

in agreement with the side length $L = 10$ of the simulation box. A partial transparency was used for the box so that the particle trajectories remain visible inside the domain. Finally, the full time evolution was inspected using the animation controls in ParaView, and the resulting visualisation can also be exported as a movie through File → Save Animation.

The visualisation confirms the expected qualitative behaviour of the simulation. The particles remain confined to the cubic domain because of the periodic boundary conditions, while their arrangement evolves dynamically due to the Lennard–Jones interaction. The colour variation among the spheres reflects the distribution of particle speeds, and the force arrows provide a direct picture of the instantaneous local interactions within the system.

References

- Course material from the Computational Statistical Physics lectures, including lecture slides and the provided skeleton Python code for the Monte Carlo simulation of the Ising model.